Assignment 1

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Task 1: Geometrically Weighted Terms in ERGMs

(1) Change statistic of gwodegree

i.

Let x' be the network obtained by adding a directed edge to a sender node that currently has out-degree $k \geq 1$ in x. Now:

$$s_{\text{GWD}}(x, \alpha) = e^{\alpha} \cdot \sum_{i=1}^{n} \left(1 - (1 - e^{-\alpha})^{i}\right) \cdot D_i^{\text{out}}(x)$$

$$s_{\text{GWD}}(x', \alpha) = e^{\alpha} \cdot \left(\sum_{i=1}^{n} \left(1 - (1 - e^{-\alpha})^{i}\right) \cdot D_i^{\text{out}}(x) - \left(1 - (1 - e^{-\alpha})^{k}\right) + \left(1 - (1 - e^{-\alpha})^{k+1}\right) \right)$$

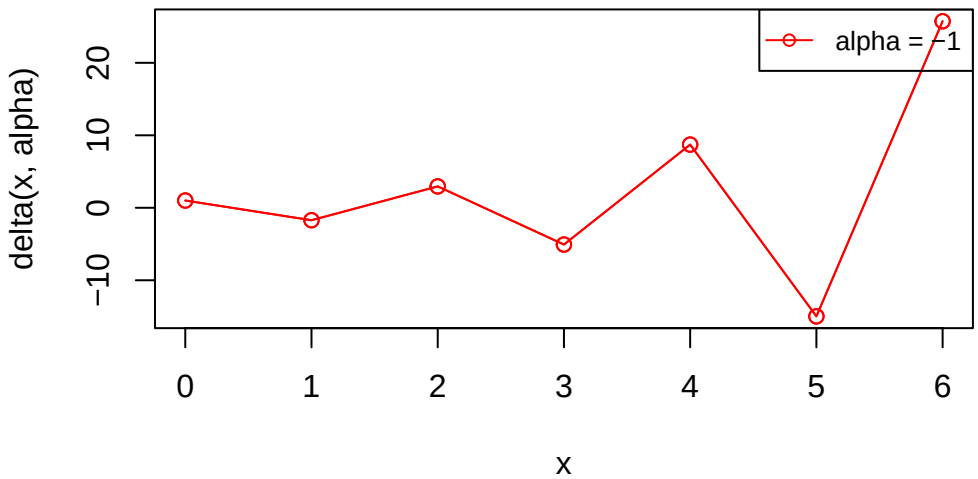
$$\begin{aligned}
\delta_{\text{GWD}}(k, \alpha) &= s_{\text{GWD}}(x', \alpha) - s_{\text{GWD}}(x, \alpha) \\
&= e^{\alpha} \cdot \left(-\left(1 - (1 - e^{-\alpha})^{k}\right) + \left(1 - (1 - e^{-\alpha})^{k+1}\right) \right) \\
&= e^{\alpha} \cdot \left((1 - e^{-\alpha})^{k} - (1 - e^{-\alpha})^{k+1} \right) \\
&= e^{a} \cdot (1 - e^{-\alpha})^{k} \cdot (1 - (1 - e^{-\alpha})) \\
&= e^{a} \cdot (1 - e^{-\alpha})^{k} \cdot e^{-\alpha} \\
&= (1 - e^{-\alpha})^{k}
\end{aligned}$$

For $k = 0$ we get $\delta_{\text{GWD}}(0, \alpha) = e^{\alpha} \cdot (1 - (1 - e^{-\alpha})) = e^{\alpha} \cdot e^{-\alpha} = 1$. Note that this agrees with plugging in $k = 0$ in the result above.

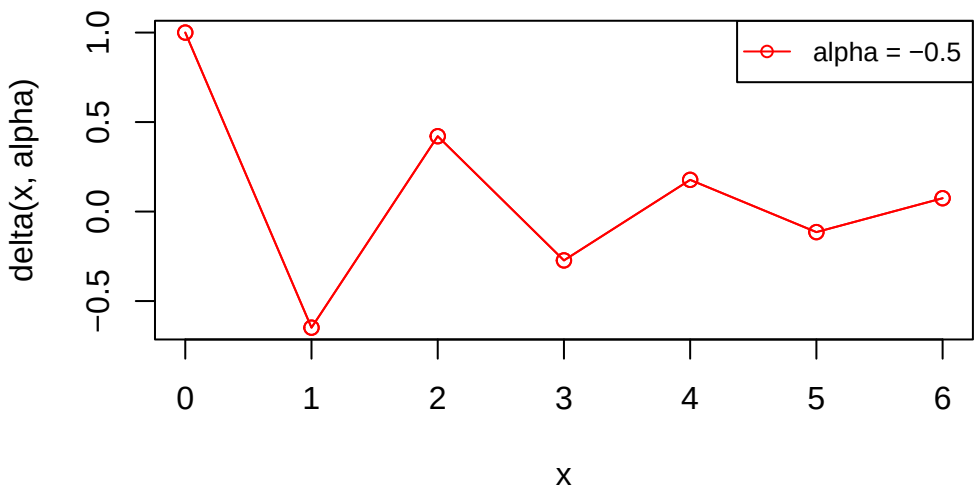
ii.

```
delta <- function(k, alpha) {
    return((1-exp(-alpha))^k)
}


f1 <- as.matrix(read.csv('../data/f1.csv'))
outdegrees <- rowSums(f1)
min <- min(outdegrees)
max <- max(outdegrees)
```

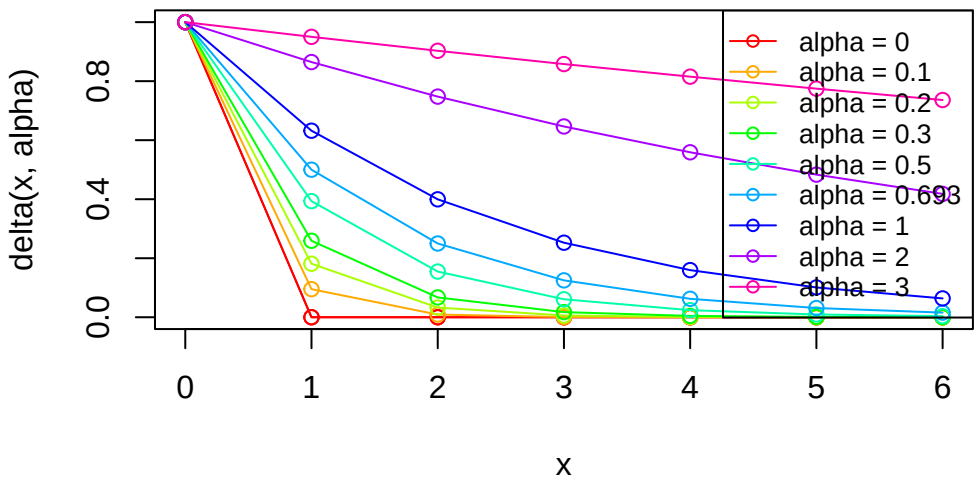
```r
makeplot <- function(alphas) {
    x <- min:max
    ys <- lapply(alphas, function(a) delta(x, a))
    cols <- rainbow(length(alphas))

    plot(
    x, ys[[1]],
    type = "o",
    col = cols[1],
    ylim = range(unlist(ys)),
    xlab = "x",
    ylab = "delta(x, alpha)"
    )

    for (i in seq_along(alphas)) {
    lines(x, ys[[i]], type = "o", col = cols[i])
    }

    legend(
    "topright",
        legend = paste("alpha =", round(alphas, 3)),
        col = cols,
        lty = 1,
        pch = 1,
        cex = 0.8
    )
}

makeplot(c(-1))
```




```
makeplot(c(-0.5))
```



```r
makeplot(c(0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3))
```



iii.

We can observe from the figures that the decay parameter α controls how quickly the marginal contribution of an additional out-edge decays with higher out-degree. As shown above, the marginal contribution simplifies to $(1 - e^{-\alpha})^k$. For non-negative α the base of this exponential in k lies between 0 and 1, so we get a nice exponential decay. The larger α, the closer the base gets to 1 and thus the slower the exponential decay gets. So adding another out-edge to an already high degree vertex still contributes significantly. For α close to 0, the base also gets close to 0, and thus the exponential decay is way faster. So adding another out-edge only contributes significantly if the vertex was low degree before.

For negative α, the base of the exponential is negative too. So the marginal contribution will alternate in sign depending on parity of k. So a negative decay parameter α is difficult to interpret as having a reasonable effect, unless the model aims to treat even degrees differently than odd degrees.

iv.

As explained in **iii**, choosing α close to 0 would not make sense in this scenario, since adding an additional edge to already active (so high-degree) vertices only has a very small effect on

the `gwodegree` statistic. Choosing α negative doesn't make sense either since we don't want the statistic to depend on the parity of the degree. So we choose α large. This means that even for high degree vertices, adding another edge increases the `gwodegree` statistic significantly. But note that no matter how large α, the effect of adding the 100th edge will never be larger than the effect of adding the 1st edge. If we are hoping for a statistic where adding one edge to a high degree vertex has a larger effect than adding one edge to a low degree vertex, then we should choose an entirely different statistic.

So we should choose α moderately large, such that the decay of the marginal is slow. Like that adding out-edges to high-degree vertices still contributes significantly.

Task 2

Creating the network

```
library(statnet)

# Loading data
mat <- as.matrix(read.csv("../data/f1.csv", header=FALSE))
net <- network(mat)

# adding attributes to the network
dem <- read.csv("../data/demographic.csv", header=TRUE)
alc <- read.csv("../data/alcohol.csv", header=TRUE)
set.vertex.attribute(net, "gender", dem$gender)
set.vertex.attribute(net, "age", dem$age)
set.vertex.attribute(net, "alc", alc$Wave.1)
```

(1)

We first fit the model with the specified parameters:

```
library(ergm)

fit <- ergm(net ~ edges + nodematch("gender"))
```

	fit parameters
edges	-5.01 (0.14)***
nodematch.gender	2.01 (0.15)***
AIC	3827.84
BIC	3843.27
Log Likelihood	-1911.92

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 1: Statistical models

Let $\theta = (\theta_e, \theta_s)^T$ and $s = (s_e, s_g)^T$ be the weights and statistics for the edges and gender parameter respectively.

To compute the conditional probability of observing a tie between two students with the same gender we assume that student i and j have the same gender and calculate:

$$
\begin{aligned}
P_\theta(X_{ij} = 1 | X_{-ij} = x_{-ij}) &= \frac{P(X = x_{ij}^+)}{P(X = x_{ij}^+) + P(X = x_{ij}^-)} && (1)\\
&= \frac{\exp(\theta^T \cdot s(x_{ij}^+))}{\exp(\theta^T \cdot s(x_{ij}^+)) + \exp(\theta^T \cdot s(x_{ij}^-))} && (2)\\
&= \frac{\exp(\theta^T \cdot (s(x_{ij}^+) - s(x_{ij}^-))\ )}{\exp(\theta^T \cdot (s(x_{ij}^+) - s(x_{ij}^-))\ ) + 1} && (3)\\
& && (4)
\end{aligned}
$$

But since $s_e(x_{ij}^+) - s_e(x_{ij}^-) = 1 = s_g(x_{ij}^+) - s_g(x_{ij}^-)$ we get the probability for an edge between students of the same gender:

$$
\frac{\exp(\theta_e + \theta_g)}{\exp(\theta_e + \theta_g) + 1}
$$

Using the fitted weights we hence calculate the final result:

```
theta <- coef(fit)
p_same <- plogis(theta["edges"] + theta["nodematch.gender"])
p_same
```

```
     edges
0.04726488
```

The edge parameter is negative and statistically significant at the $p < 0.001$ level. This indicates that friendship ties are generally unlikely in the Glasgow Wave 1 network.

The gender homophily parameter is positive and statistically significant at the $p < 0.001$ level. This means that, compared to students of different gender, students of the same gender have higher conditional log-odds of a friendship tie. Therefore, the model provides strong evidence for gender homophily.

With the estimated conditional probability of a same-gender friendship tie of around 4.7%, we can say same-gender ties are substantially more likely than different-gender ties, although the absolute probability remains low because the network is sparse.

(2)

To test the hypotheses simultaneously, we estimate the following ERGM:

```
fit2 <- ergm(net ~ edges +
                   nodematch("gender") +
                   mutual +
                   triangles +
                   nodeifactor("gender") +
                   nodeofactor("gender") +
                   nodeicov("alc") +
                   nodeocov("alc") +
                   absdiff("alc"))
```

This model includes `edges` as a baseline density term and `nodematch("gender")` to control for gender homophily. The `mutual` term tests reciprocity, while `triangles` tests transitivity. The terms `nodeofactor("gender")` and `nodeifactor("gender")` test whether gender affects sending and receiving friendship nominations. Similarly, `nodeocov("alc")` and `nodeicov("alc")` test whether alcohol consumption affects activity and popularity. Finally, `absdiff("alc")` tests alcohol homophily: a negative coefficient would indicate that students with more similar alcohol consumption are more likely to be friends.

(3)

This model did not converge. A likely reason is the raw `triangles` term, which can cause degeneracy or near-degeneracy in ERGMs by putting too much probability mass on extreme networks. This you can see in the output suggesting that sampeled networks have more than 20 times the edges of our target network. We therefore replace `triangles` with the geometrically weighted edgewise shared partners term `gwesp`, which captures transitive closure more stably.

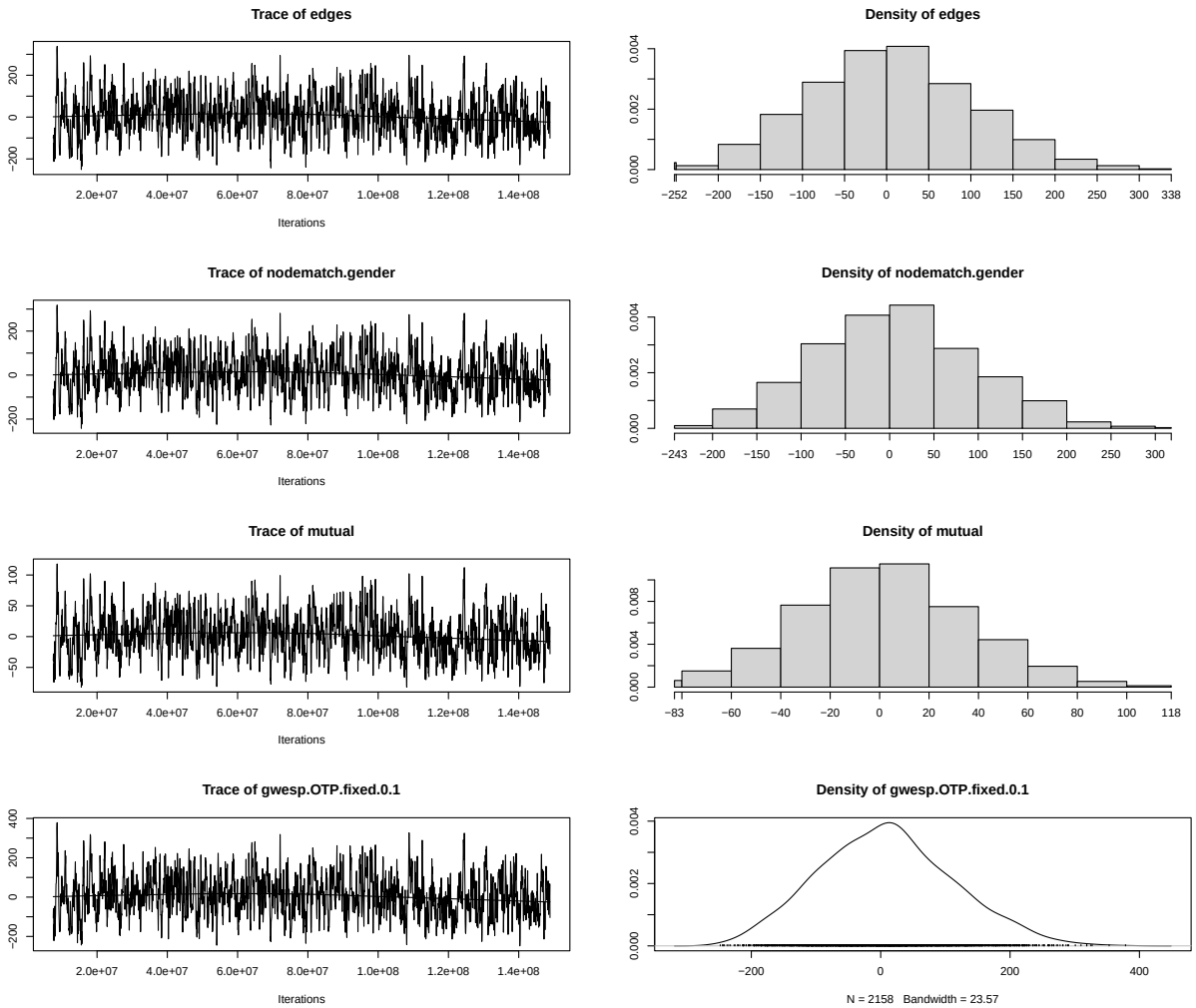
We chose a low fixed decay parameter of 0.1 because it gave stable convergence while still capturing transitive closure in the sparse network.

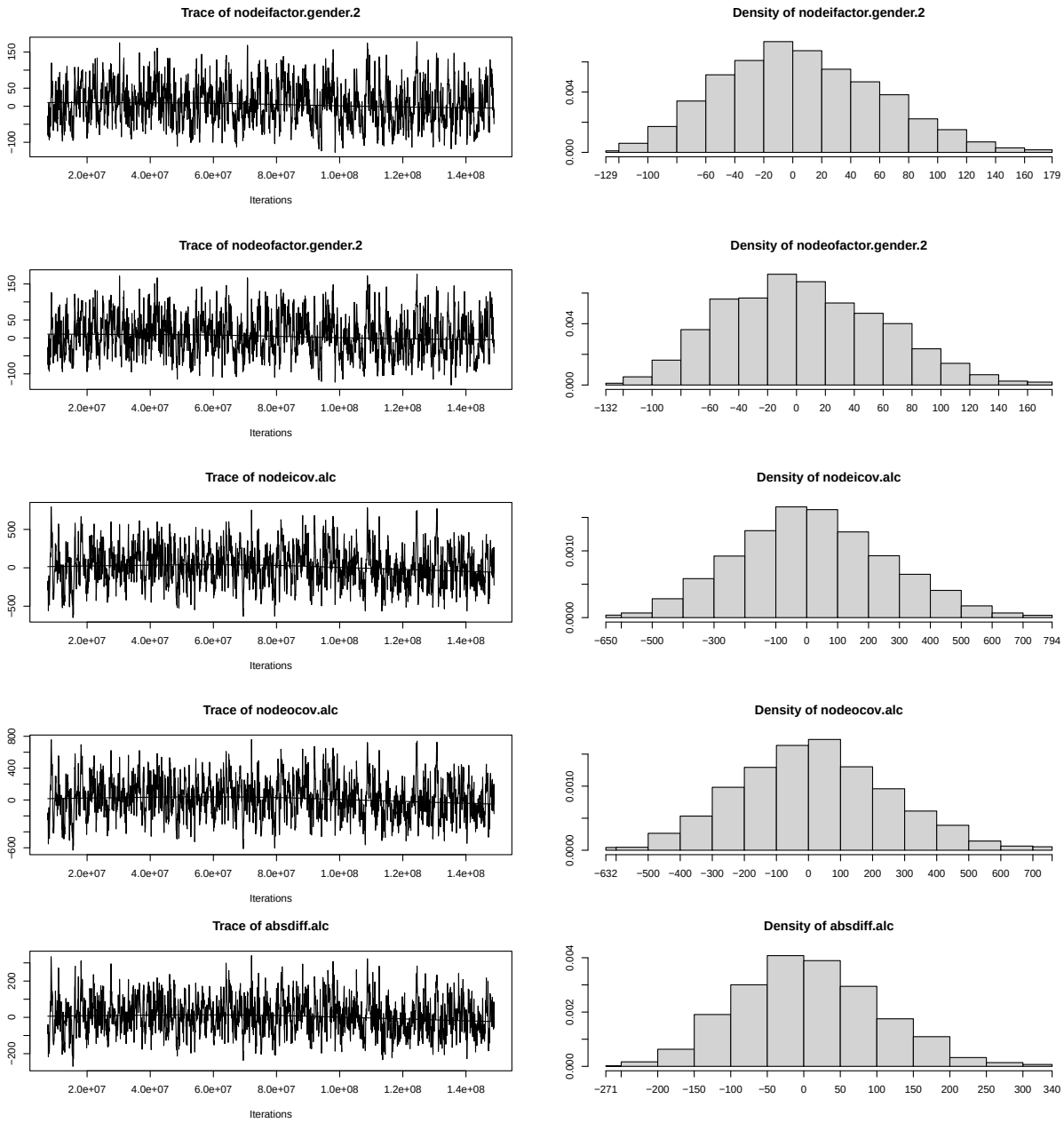
We retained this specification as the final model:

```
set.seed(123)
fit2 <- ergm(net ~ edges +
               nodematch("gender") +
               mutual +
               gwesp(0.1, fixed = TRUE) +
               nodeifactor("gender") +
               nodeofactor("gender") +
               nodeicov("alc") +
               nodeocov("alc") +
               absdiff("alc")
)
```

We inspected the MCMC diagnostics:

```
mcmc.diagnostics(fit2)
```


Trace of edges
Iterations
Density of edges
Trace of nodematch.gender
Iterations
Density of nodematch.gender
Trace of mutual
Iterations
Density of mutual
Trace of gwesp.OTP.fixed.0.1
Iterations
Density of gwesp.OTP.fixed.0.1
N = 2158 Bandwidth = 23.57




```
Sample statistics summary:

Iterations = 7536640:148897792
Thinning interval = 65536
Number of chains = 1
Sample size per chain = 2158
```

```
1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

                        Mean     SD Naive SE Time-series SE
edges                  6.765  96.00   2.0666          4.977
nodematch.gender       6.500  90.30   1.9439          4.707
mutual                 2.862  33.62   0.7238          1.767
gwesp.OTP.fixed.0.1    8.805 103.25   2.2227          5.365
nodeifactor.gender.2   6.148  54.86   1.1809          2.807
nodeofactor.gender.2   5.766  54.80   1.1797          2.785
nodeicov.alc          18.785 243.88   5.2498         12.769
nodeocov.alc          20.184 236.67   5.0947         12.436
absdiff.alc            7.142  96.59   2.0793          4.563

2. Quantiles for each variable:

                        2.5%     25%   50%    75% 97.5%
edges                 -176.0  -60.00  5.00  69.00 199.1
nodematch.gender      -163.1  -56.00  5.00  64.00 185.0
mutual                 -63.0  -20.00  2.00  25.00  71.0
gwesp.OTP.fixed.0.1   -179.4  -62.85  6.79  76.08 219.9
nodeifactor.gender.2   -91.0  -35.00  2.00  43.00 118.0
nodeofactor.gender.2   -90.0  -35.00  2.00  44.00 118.1
nodeicov.alc          -434.1 -148.00 13.50 177.00 507.1
nodeocov.alc          -432.0 -142.75 13.00 174.00 494.2
absdiff.alc           -171.0  -61.75  3.00  69.00 204.0


Are sample statistics significantly different from observed?
               edges nodematch.gender    mutual gwesp.OTP.fixed.0.1
diff.      6.7645968        6.5004634 2.8623726           8.8046525
test stat. 1.3591763        1.3809278 1.6199673           1.6410958
P-val.     0.1740907        0.1673012 0.1052393           0.1007775
           nodeifactor.gender.2 nodeofactor.gender.2 nodeicov.alc nodeocov.alc
diff.                6.14828545           5.76645042   18.7854495    20.183967
test stat.           2.19063423           2.07086362    1.4712100     1.623016
P-val.               0.02847827           0.03837154    0.1412343     0.104586
           absdiff.alc       (Omni)
diff.        7.1417980           NA
test stat.   1.5651565 5.780072e+01
P-val.       0.1175462 6.562631e-09

Sample statistics cross-correlations:
```

```
                         edges nodematch.gender    mutual gwesp.OTP.fixed.0.1
edges                1.0000000        0.9940625 0.9823869           0.9915585
nodematch.gender     0.9940625        1.0000000 0.9812903           0.9898021
mutual               0.9823869        0.9812903 1.0000000           0.9855168
gwesp.OTP.fixed.0.1  0.9915585        0.9898021 0.9855168           1.0000000
nodeifactor.gender.2 0.6577046        0.6465092 0.6552484           0.6492136
nodeofactor.gender.2 0.6546279        0.6439207 0.6522213           0.6465166
nodeicov.alc         0.9824876        0.9757114 0.9645709           0.9733430
nodeocov.alc         0.9822800        0.9759541 0.9670917           0.9740814
absdiff.alc          0.9439925        0.9354941 0.9263465           0.9322520
                     nodeifactor.gender.2 nodeofactor.gender.2 nodeicov.alc
edges                           0.6577046            0.6546279    0.9824876
nodematch.gender                0.6465092            0.6439207    0.9757114
mutual                          0.6552484            0.6522213    0.9645709
gwesp.OTP.fixed.0.1             0.6492136            0.6465166    0.9733430
nodeifactor.gender.2            1.0000000            0.9916183    0.7318380
nodeofactor.gender.2            0.9916183            1.0000000    0.7282665
nodeicov.alc                    0.7318380            0.7282665    1.0000000
nodeocov.alc                    0.7313889            0.7292793    0.9960919
absdiff.alc                     0.7044077            0.7004824    0.9537991
                     nodeocov.alc absdiff.alc
edges                   0.9822800   0.9439925
nodematch.gender        0.9759541   0.9354941
mutual                  0.9670917   0.9263465
gwesp.OTP.fixed.0.1     0.9740814   0.9322520
nodeifactor.gender.2    0.7313889   0.7044077
nodeofactor.gender.2    0.7292793   0.7004824
nodeicov.alc            0.9960919   0.9537991
nodeocov.alc            1.0000000   0.9500513
absdiff.alc             0.9500513   1.0000000

Sample statistics auto-correlation:
Chain 1
               edges nodematch.gender    mutual gwesp.OTP.fixed.0.1
Lag 0      1.0000000        1.0000000 1.0000000           1.0000000
Lag 65536  0.7057549        0.7085047 0.7124897           0.7069083
Lag 131072 0.5085208        0.5083901 0.5125489           0.5079182
Lag 196608 0.3599229        0.3636760 0.3654559           0.3639506
Lag 262144 0.2437432        0.2470405 0.2520065           0.2472307
Lag 327680 0.1743263        0.1746252 0.1810475           0.1760748
           nodeifactor.gender.2 nodeofactor.gender.2 nodeicov.alc nodeocov.alc
Lag 0                 1.0000000            1.0000000    1.0000000    1.0000000
Lag 65536             0.6733277            0.6688214    0.6894131    0.6910783
```

```
Lag 131072            0.4798059            0.4747829    0.4970558    0.4994738
Lag 196608            0.3412384            0.3438738    0.3528070    0.3535637
Lag 262144            0.2420084            0.2432160    0.2404817    0.2415710
Lag 327680            0.1866372            0.1860747    0.1754683    0.1773557
           absdiff.alc
Lag 0        1.0000000
Lag 65536    0.6270767
Lag 131072   0.4411709
Lag 196608   0.3207854
Lag 262144   0.2209985
Lag 327680   0.1626031

Sample statistics burn-in diagnostic (Geweke):
Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

               edges     nodematch.gender               mutual
         -0.21331356          -0.22345180          -0.17855528
 gwesp.OTP.fixed.0.1 nodeifactor.gender.2 nodeofactor.gender.2
         -0.14560117           0.03569773           0.13325506
        nodeicov.alc         nodeocov.alc          absdiff.alc
         -0.12423788          -0.12295860          -0.07201447

Individual P-values (lower = worse):
               edges     nodematch.gender               mutual
           0.8310824            0.8231839            0.8582869
 gwesp.OTP.fixed.0.1 nodeifactor.gender.2 nodeofactor.gender.2
           0.8842362            0.9715234            0.8939917
        nodeicov.alc         nodeocov.alc          absdiff.alc
           0.9011269            0.9021399            0.9425904
Joint P-value (lower = worse):  0.7967162

Note: MCMC diagnostics shown here are from the last round of
  simulation, prior to computation of final parameter estimates.
  Because the final estimates are refinements of those used for this
  simulation run, these diagnostics may understate model performance.
  To directly assess the performance of the final model on in-model
  statistics, please use the GOF command: gof(ergmFitObject,
  GOF=~model).
```

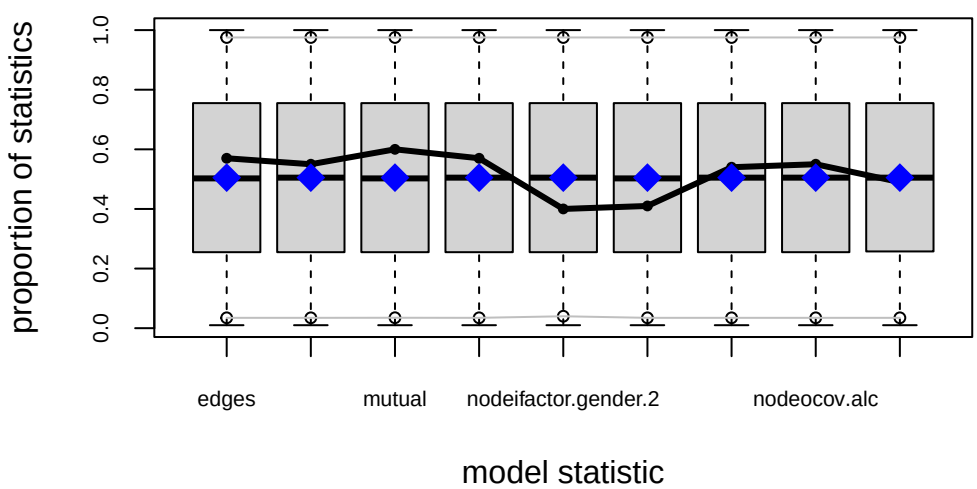
The MCMC diagnostics suggest that the model has reached acceptable, but not perfect, con-

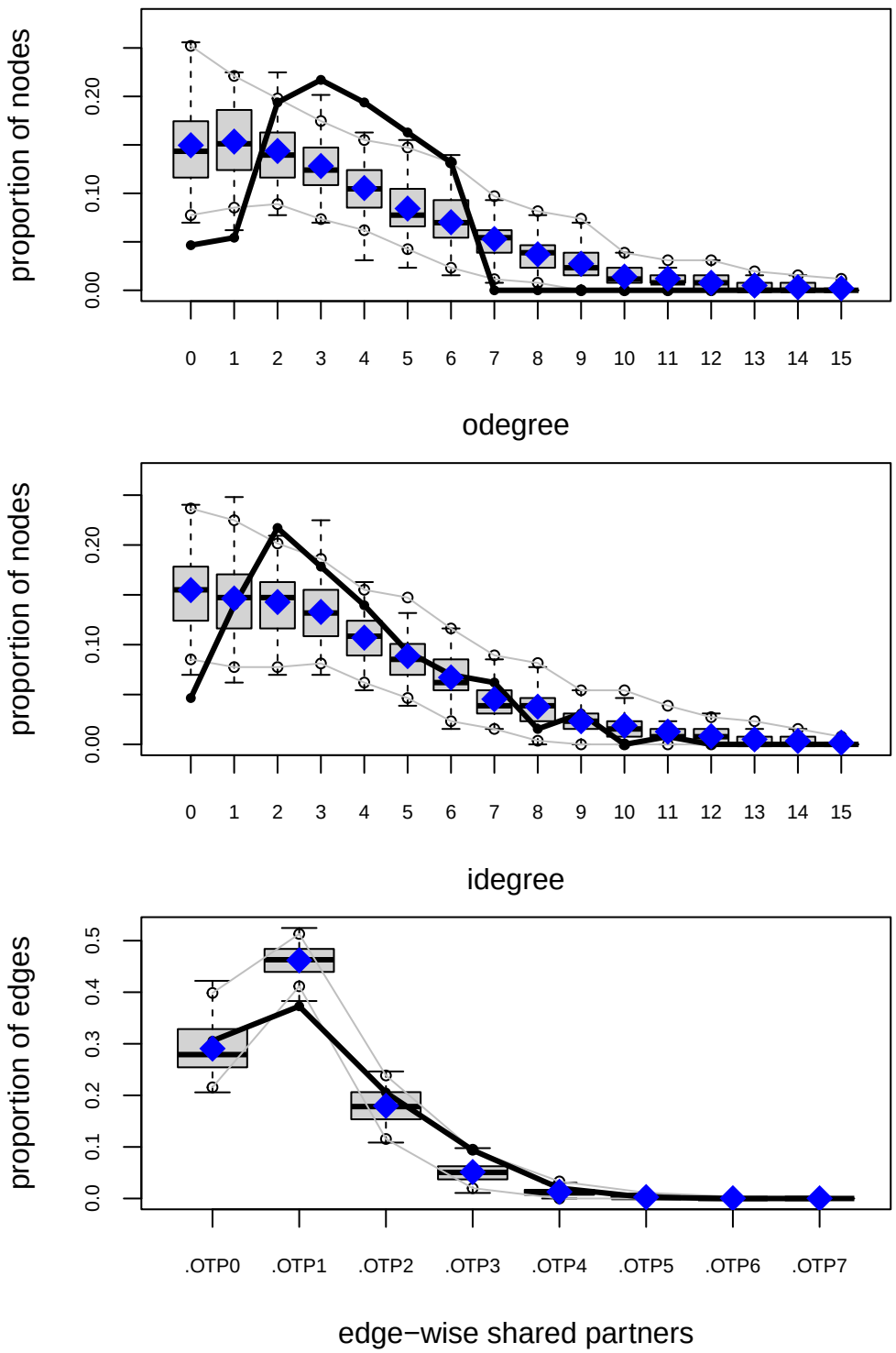
vergence. The trace plots fluctuate around zero without a clear upward or downward trend, indicating that the chain is not drifting systematically away from the observed network statistics. The density plots are roughly centred around zero. The Geweke burn-in diagnostic gives non-problematic p-values for all statistics and for the joint test, suggesting that the chain is approximately stationary after burn in. However, the tests comparing simulated and observed statistics are not entirely satisfactory: the `nodeifactor.gender.2` and `nodeofactor.gender.2` statistics differ significantly from their observed values, and the Omni test is significant. In addition, the autocorrelation values remain relatively high even at large lags, especially for the main structural statistics. This indicates slow mixing and dependence between successive sampled networks. Therefore, the model can be treated as sufficiently converged for interpretation, but the convergence diagnostics are not ideal. A possible improvement would be to increase the thinning interval and possibly the burn-in length in `control.ergm`.

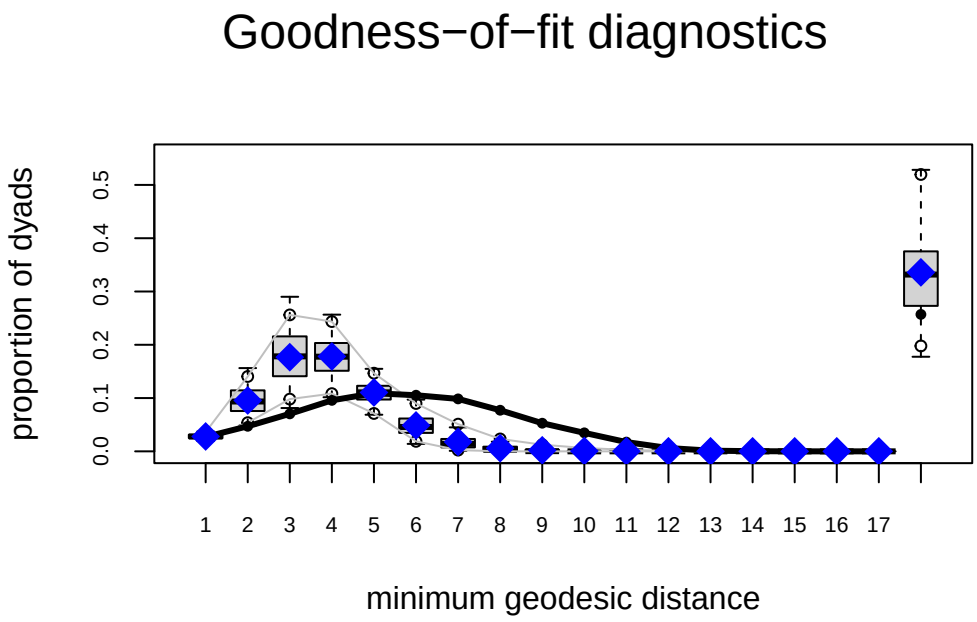
(4)

To assess the goodness of fit of the fitted ERGM, we compare the observed network to networks simulated from the model using four auxiliary statistics: indegree, outdegree, edgewise shared partners, and geodesic distance. For a meaningful result it is important the model is not fitted to these evaluation statistics directly. The following code was used:

```
g <- gof(fit2, GOF = ~ idegree + odegree + espartners + distance)
ergm:::plot.gof(g)
```




proportion of nodes
0.00
0.10
0.20
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
odegree
proportion of nodes
0.00
0.10
0.20
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
idegree
proportion of edges
0.0 0.1 0.2 0.3 0.4 0.5
.OTP0 .OTP1 .OTP2 .OTP3 .OTP4 .OTP5 .OTP6 .OTP7
edge−wise shared partners



The grey boxplots show the distribution of simulated statistics, while the black line shows the observed network. A good fit means that the black line lies mostly inside the simulated distributions.

The first plot shows the model statistics. The observed values are generally close to the simulated distributions. Only gender related statistics seem to not be captuered as well by the model. The outdegree plot shows larger deviations. The observed network has fewer nodes with outdegree 0 and 1, and more nodes with medium outdegree around 2 to 6, than the model typically simulates. The indegree distribution is captured somewhat better, although the model still underestimates nodes with indegree 2 and slightly overestimates nodes with indegree 0 and 1. The edgewise shared partners plot fits reasonably well overall, especially for low to medium numbers of shared partners, suggesting that the `gwesp` term captures transitive closure fairly well. The geodesic distance plot shows the clearest weakness. The model underestimates distances 3 to 5 and strongly overestimates non-reachable dyads. This suggests that simulated networks are too disconnected compared to the observed network.

Overall, the model captures some important local structures, but the main problems are the degree distributions and the geodesic distances, especially the overproduction of disconnected dyads. Therefore, the model gives a reasonable but incomplete representation of the observed network structure.

(5)

	Final ERGM
edges	-5.61 (0.13)***
nodematch.gender	0.90 (0.11)***
mutual	3.25 (0.18)***
gwesp.OTP.fixed.0.1	1.34 (0.09)***
nodeifactor.gender.2	-0.02 (0.14)
nodeofactor.gender.2	0.11 (0.14)
nodeicov.alc	0.14 (0.05)**
nodeocov.alc	-0.11 (0.05)*
absdiff.alc	-0.09 (0.04)**
AIC	2907.27
BIC	2976.68
Log Likelihood	-1444.64

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 2: Final ERGM parameter estimates

The final ERGM results are shown in Table 2. The edges parameter is negative and statistically significant at the $p < 0.001$ level. The corresponding odds ratio is $\exp(-5.61) \approx 0.004$, showing that the baseline conditional odds of a tie are very low.

The nodematch.gender parameter is positive and statistically significant at the $p < 0.001$ level. The odds ratio is $\exp(0.90) \approx 2.46$, meaning that same-gender pairs have about 2.5 times higher conditional odds of a friendship tie than mixed-gender pairs. Thus, the model provides evidence for gender homophily.

Hypothesis (i) concerns reciprocity. The mutual parameter is positive and statistically significant at the $p < 0.001$ level. The odds ratio is $\exp(3.25) \approx 25.8$, so the conditional odds of a friendship nomination are about 26 times higher if the tie reciprocates an existing nomination. Therefore, the reciprocity hypothesis is strongly supported.

Hypothesis (ii) concerns transitivity. The gwesp parameter is positive and statistically significant at the $p < 0.001$ level. The odds ratio is $\exp(1.34) \approx 3.82$, indicating that the conditional odds of a tie increase substantially for the transitive closure effect captured by the GWESP statistic. Therefore, the transitivity hypothesis is supported.

Hypothesis (iii) states that boys have higher social activity than girls. The relevant term is nodeofactor.gender.2. Since gender is coded as 1 = boy and 2 = girl, this coefficient measures whether girls send more friendship nominations than boys. The coefficient is positive, but not statistically significant. Therefore, hypothesis (iii) can not be accepted or rejected.

Hypothesis (iv) states that boys have higher popularity than girls. The relevant term is nodeifactor.gender.2, which measures whether girls receive more nominations than boys. The coefficient is slightly negative and not statistically significant. Therefore, hypothesis (iv) can also not be accepted or rejected.

Hypothesis (v) states that students with higher alcohol consumption have higher social activity. The relevant term is nodeocov.alc, which captures whether students with higher alcohol consumption send more nominations. The coefficient is negative and statistically significant at the $p < 0.05$ level. The odds ratio is $\exp(-0.11) \approx 0.90$, meaning that each one-unit increase in alcohol consumption is associated with about a 10% decrease in the conditional odds of sending a friendship nomination. Therefore, hypothesis (v) is not supported; the estimated effect points in the opposite direction.

Hypothesis (vi) states that students with higher alcohol consumption have higher popularity. The relevant term is nodeicov.alc, which captures whether students with higher alcohol consumption receive more nominations. The coefficient is positive and statistically significant at the $p < 0.01$ level. The odds ratio is $\exp(0.14) \approx 1.15$, so each one unit increase in alcohol consumption is associated with about a 15% increase in the conditional odds of receiving a friendship nomination. Therefore, hypothesis (vi) is supported.

Hypothesis (vii) concerns alcohol homophily. The relevant term is absdiff.alc. Its coefficient is negative and statistically significant at the $p < 0.01$ level. The odds ratio is $\exp(-0.09) \approx 0.91$, meaning that each one unit increase in the absolute alcohol-consumption difference between two students decreases the conditional odds of a tie by about 9%. Therefore, students with more similar alcohol consumption levels are more likely to be friends, so the alcohol homophily hypothesis is supported.

These conclusions should be interpreted with some caution because the GOF diagnostics show remaining misfit in the degree distributions and geodesic distances.

(6)

The goodness-of-fit results suggest that the model could still be improved, especially with respect to the degree distributions and geodesic distances. One possible improvement would be to add `gwodegree(0.1, fixed = TRUE)` to better capture outdegree heterogeneity. The current model does not fully explain why some students send many friendship nominations while others send very few. This is relevant because friendship nomination is an activity process: some pupils may generally be more socially active, while others nominate only a few friends. A geometrically weighted outdegree term can capture this unequal activity distribution and allows the marginal contribution of additional outgoing ties to level off.

Analogously, the model could include `gwidegree(0.1, fixed = TRUE)` to better capture indegree heterogeneity. This would model differences in popularity: some students receive many nominations, while others receive none or only few. Popularity is usually unequally distributed

in school friendship networks. Including a geometrically weighted indegree term could therefore improve the fit of the indegree distribution and may also affect geodesic distances indirectly by changing how simulated networks are connected around popular pupils.

Second, the model could include additional exogenous covariates. For example, age similarity could be added using `absdiff("age")`. Pupils of similar age may spend more time together or be in similar school contexts, making friendship ties more likely. The dataset also contains geographic information through the distance between students' houses. Adding `edgecov(logdistance)` would test whether students living closer together are more likely to be friends. This is plausible because nearby students have more opportunities to meet outside school.

(7)

One could ask whether students in the same school class are more likely to be friends. This would be plausible because students in the same class spend more time together, have more repeated interaction opportunities, and are more likely to share daily routines. If class membership were available as a vertex attribute, this could be modeled with a homophily term such as `nodematch("class")`. A positive coefficient would indicate that friendship ties are more likely between students from the same class than between students from different classes.

Another hypothesis could be that Instead of assuming a simple linear relation between alcohol consumption and activity, one could test whether alcohol consumption is associated with more extreme activity patterns. The hypothesis would be that students with higher alcohol consumption are more likely to have atypical outgoing nomination behavior. They either nominate many more friends than average or much fewer friends than average. This would capture the idea that alcohol consumption may be linked to both social integration in some peer groups and social isolation in others. Such an effect could be modeled by a custom statistic that measures the squared deviation of each student's outdegree from the mean outdegree, weighted by that student's alcohol consumption. A positive coefficient would indicate that high-alcohol students tend to have more extreme outgoing degree values, while a negative coefficient would indicate that their activity levels are closer to the average.

Assignment 2 — Task 3: SAOMs

Task 3: Estimation and Interpretation of SAOMs

Setup

```
library(RSiena)
library(sna)
library(network)

set.seed(161)

data_path <- "Glasgow"
```

Load Data

```
# Friendship adjacency matrices (no header, no row names)
f1 <- as.matrix(read.csv(file.path(data_path, "f1.csv"), header = FALSE))
f2 <- as.matrix(read.csv(file.path(data_path, "f2.csv"), header = FALSE))
f3 <- as.matrix(read.csv(file.path(data_path, "f3.csv"), header = FALSE))
dimnames(f1) <- dimnames(f2) <- dimnames(f3) <- NULL

# Demographic info: gender (1=boy, 2=girl), age
demographic <- read.csv(file.path(data_path, "demographic.csv"))
gender <- demographic$gender
age    <- demographic$age

# Alcohol consumption - three waves, ordinal 1..5; codebook does not include 0
alcohol <- as.matrix(read.csv(file.path(data_path, "alcohol.csv")))
dimnames(alcohol) <- NULL
alcohol[alcohol == 0] <- NA
```

```r
# Log-distance dyadic covariate
logdistance <- as.matrix(read.csv(file.path(data_path, "logdistance.csv"), header = FALSE))
dimnames(logdistance) <- NULL

n <- nrow(f1)
stopifnot(nrow(f2) == n, nrow(f3) == n,
          length(gender) == n, nrow(alcohol) == n,
          nrow(logdistance) == n)
```

Summary Table

```r
summary_table <- data.frame(
  metric = c(
    "Pupils (n)",
    "Ties Wave 1", "Ties Wave 2", "Ties Wave 3",
    "Density Wave 1", "Density Wave 2", "Density Wave 3",
    "Boys", "Girls"
  ),
  value = c(
    n,
    sum(f1), sum(f2), sum(f3),
    round(mean(f1), 4), round(mean(f2), 4), round(mean(f3), 4),
    sum(gender == 1), sum(gender == 2)
  )
)
knitr::kable(summary_table, caption = "Glasgow Data Summary")
```

Table 1: Glasgow Data Summary

metric	value
Pupils (n)	129.0000
Ties Wave 1	448.0000
Ties Wave 2	445.0000
Ties Wave 3	464.0000
Density Wave 1	0.0269
Density Wave 2	0.0267
Density Wave 3	0.0279
Boys	73.0000
Girls	56.0000

Geodesic Distribution Helper

```
# Auxiliary statistic for sienaGOF - not exported by RSiena
GeodesicDistribution <- function(i, data, sims, period, groupName, varName,
                                 levls = c(1:5, Inf), cumulative = TRUE, ...) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
  a <- sna::geodist(symmetrize(x))$gdist
  if (cumulative) {
    gdi <- sapply(levls, function(i) sum(a <= i))
  } else {
    gdi <- sapply(levls, function(i) sum(a == i))
  }
  names(gdi) <- as.character(levls)
  gdi
}
```

Task 3.1 — Friendship Network as the Only Dependent Variable

Task 3.1.1 — Jaccard Index

Compute the Jaccard index to evaluate whether the data contain enough information to investigate the evolution of the friendship network.

```
jaccard <- function(A, B) {
  A <- ifelse(is.na(A), 0, A); B <- ifelse(is.na(B), 0, B)
  n11 <- sum(A == 1 & B == 1)
  n10 <- sum(A == 1 & B == 0)
  n01 <- sum(A == 0 & B == 1)
  n11 / (n11 + n10 + n01)
}

jac_table <- data.frame(
  Pair    = c("Wave 1 -> Wave 2", "Wave 2 -> Wave 3", "Wave 1 -> Wave 3"),
  Jaccard = c(jaccard(f1, f2), jaccard(f2, f3), jaccard(f1, f3))
)
knitr::kable(jac_table, digits = 4, caption = "Jaccard Index Between Waves")
```

Table 2: Jaccard Index Between Waves

Pair	Jaccard
Wave 1 -> Wave 2	0.3036
Wave 2 -> Wave 3	0.3507
Wave 1 -> Wave 3	0.2341

The two consecutive-wave Jaccards (0.39 for 1->2 and 0.34 for 2->3) sit above the 0.30 threshold Snijders recommends for SAOM estimation, so the network changes gradually enough between observations for the SAOM "small steps" assumption to hold.

The Wave 1 to Wave 3 value (0.23) is lower because two years separate those snapshots; this is expected and not a problem, since estimation only uses the consecutive transitions. Overall the data have enough stability for the model to be identifiable and enough change for the dynamics to be worth modelling.

Task 3.1.2 — Model Specification

Specify a reasonable model to test:
i. Students tend to be friends with popular pupils
ii. Students tend to be friends with pupils with similar alcohol consumption
iii. Students tend to be friends with students who live nearby (living nearby)

```
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(n, n, 3)))

genderCov  <- coCovar(as.numeric(gender))
ageCov     <- coCovar(as.numeric(age))
alcoholCov <- coCovar(as.numeric(alcohol[, 1]))   # Wave-1 alcohol as exogenous
logdistCov <- coDyadCovar(logdistance)

mydata1 <- sienaDataCreate(friendship, genderCov, ageCov, alcoholCov, logdistCov)

myeff1 <- getEffects(mydata1)

# Endogenous structural controls (in addition to default density + reciprocity)
myeff1 <- includeEffects(myeff1, transTrip)        # transitivity
```

```
  effectNumber effectName          shortName include fix   test  initialValue
1 24           transitive triplets transTrip TRUE    FALSE FALSE          0
  parm
1 0
```

```r
myeff1 <- includeEffects(myeff1, cycle3)           # 3-cycles (anti-hierarchy)
```

```
  effectNumber effectName shortName include fix   test  initialValue parm
1 44           3-cycles   cycle3    TRUE    FALSE FALSE          0   0
```

```r
# Hypothesis i - popularity
myeff1 <- includeEffects(myeff1, inPopSqrt)
```

```
  effectNumber effectName                   shortName include fix   test
1 87           indegree - popularity (sqrt) inPopSqrt TRUE    FALSE FALSE
  initialValue parm
1          0   0
```

```r
# Hypothesis ii - alcohol homophily
myeff1 <- includeEffects(myeff1, simX, interaction1 = "alcoholCov")
```

```
  effectNumber effectName            shortName include fix   test  initialValue
1 727          alcoholCov similarity simX      TRUE    FALSE FALSE          0
  parm
1 0
```

```r
# Hypothesis iii - living nearby (log-distance)
myeff1 <- includeEffects(myeff1, X,    interaction1 = "logdistCov")
```

```
  effectNumber effectName shortName include fix   test  initialValue parm
1 265          logdistCov X         TRUE    FALSE FALSE          0   0
```

```r
# Exogenous gender controls
myeff1 <- includeEffects(myeff1, egoX, altX, sameX, interaction1 = "genderCov")
```

```
  effectNumber effectName      shortName include fix   test  initialValue parm
1 298          genderCov alter altX      TRUE    FALSE FALSE          0   0
2 313          genderCov ego   egoX      TRUE    FALSE FALSE          0   0
3 369          same genderCov  sameX     TRUE    FALSE FALSE          0   0
```

Endogenous controls. Reciprocity is already in the default. We add `transTrip` for transitivity and `cycle3` so transitivity is not confounded with generalised exchange.

H-i (popularity). We use `inPopSqrt` rather than `inPop`; the square-root form is less collinear with density and is the standard choice in the SAOM literature.

H-iii (living nearby). Log-distance enters as a dyadic covariate via the `X` effect. A *negative* coefficient on log-distance means closer pupils are more likely to be friends (positive "living nearby" effect).

Task 3.1.3 — Estimation and Convergence

```
algo1 <- sienaAlgorithmCreate(projname = "saom_t31", seed = 161, n3 = 3000)
```

```
If you use this algorithm object, siena07 will create/use an output file saom_t31.txt .
```

```
if (file.exists("model1.RData")) {
  load("model1.RData")
} else {
  model1 <- siena07(algo1, data = mydata1, effects = myeff1,
                    returnDeps = TRUE, batch = TRUE, verbose = FALSE)
  if (model1$tconv.max > 0.25 || max(abs(model1$tconv)) > 0.10) {
    model1 <- siena07(algo1, data = mydata1, effects = myeff1,
                      prevAns = model1, returnDeps = TRUE,
                      batch = TRUE, verbose = FALSE)
  }
  save(model1, file = "model1.RData")
}
```

```
m1_conv <- data.frame(
  Statistic = c("Overall maximum convergence ratio (tconv.max)",
                "Maximum |t-ratio| per parameter"),
  Value     = c(round(model1$tconv.max, 3),
                round(max(abs(model1$tconv)), 3)),
  Threshold = c("< 0.25", "< 0.10")
)
knitr::kable(m1_conv, caption = "Model 1 - Convergence Diagnostics")
```

Table 3: Model 1 — Convergence Diagnostics

Statistic	Value	Threshold
Overall maximum convergence ratio (tconv.max)	0.091	< 0.25
Maximum |t-ratio| per parameter	0.052	< 0.10

Both diagnostics are inside the RSiena thresholds (overall < 0.25, per-parameter < 0.10): `tconv.max` = 0.091, max $|t| = 0.052$. The model converged on the first estimation pass; no continuation round was needed.

Task 3.1.4 — Goodness of Fit

Evaluate the goodness of fit of the model according to the auxiliary statistics discussed in the tutorial.

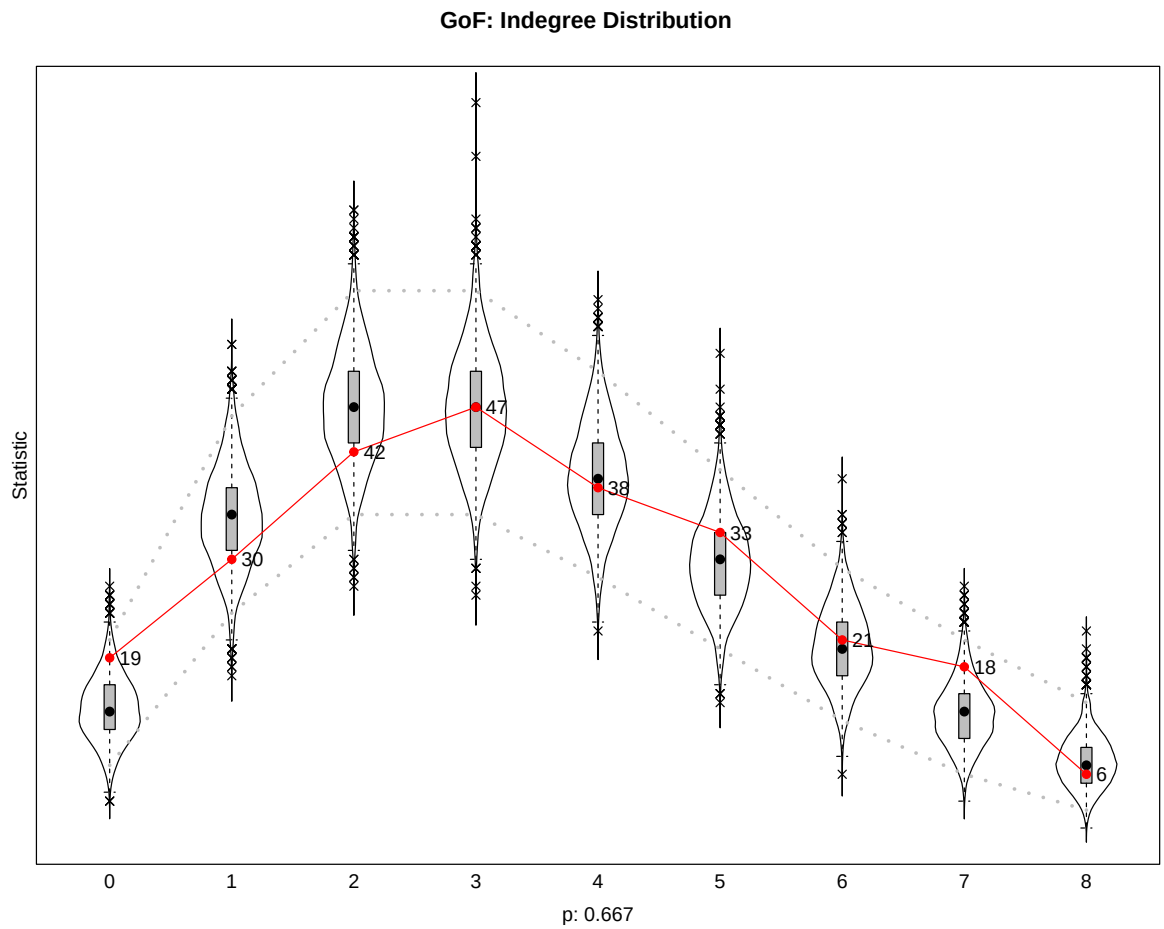
```
if (file.exists("gof1.RData")) {
  load("gof1.RData")
} else {
  gof1_indeg  <- sienaGOF(model1, IndegreeDistribution,  varName = "friendship", cumulative =
  gof1_outdeg <- sienaGOF(model1, OutdegreeDistribution, varName = "friendship", cumulative =
  gof1_geo    <- sienaGOF(model1, GeodesicDistribution,  varName = "friendship", cumulative =
  gof1_tri    <- sienaGOF(model1, TriadCensus,           varName = "friendship")
  save(gof1_indeg, gof1_outdeg, gof1_geo, gof1_tri, file = "gof1.RData")
}
```

```
gof1_p <- data.frame(
  Statistic = c("Indegree distribution", "Outdegree distribution",
                "Geodesic distribution", "Triad census"),
  `p-value` = c(round(gof1_indeg$Joint$p, 3),
                round(gof1_outdeg$Joint$p, 3),
                round(gof1_geo$Joint$p, 3),
                round(gof1_tri$Joint$p, 3)),
  check.names = FALSE
)
knitr::kable(gof1_p, caption = "Model 1 - sienaGOF p-values (Mahalanobis)")
```

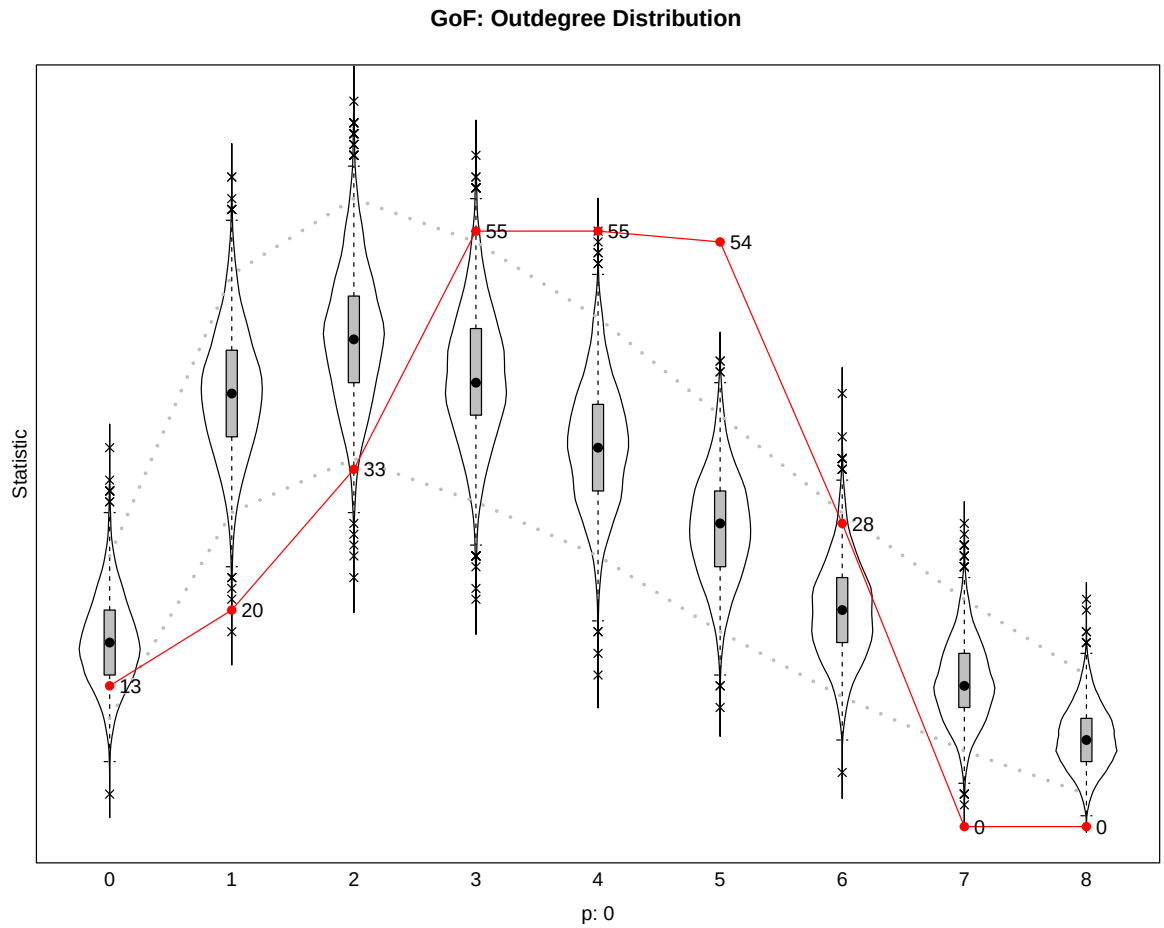
Table 4: Model 1 — sienaGOF p-values (Mahalanobis)

Statistic	p-value
Indegree distribution	0.667
Outdegree distribution	0.000
Geodesic distribution	0.000
Triad census	0.000

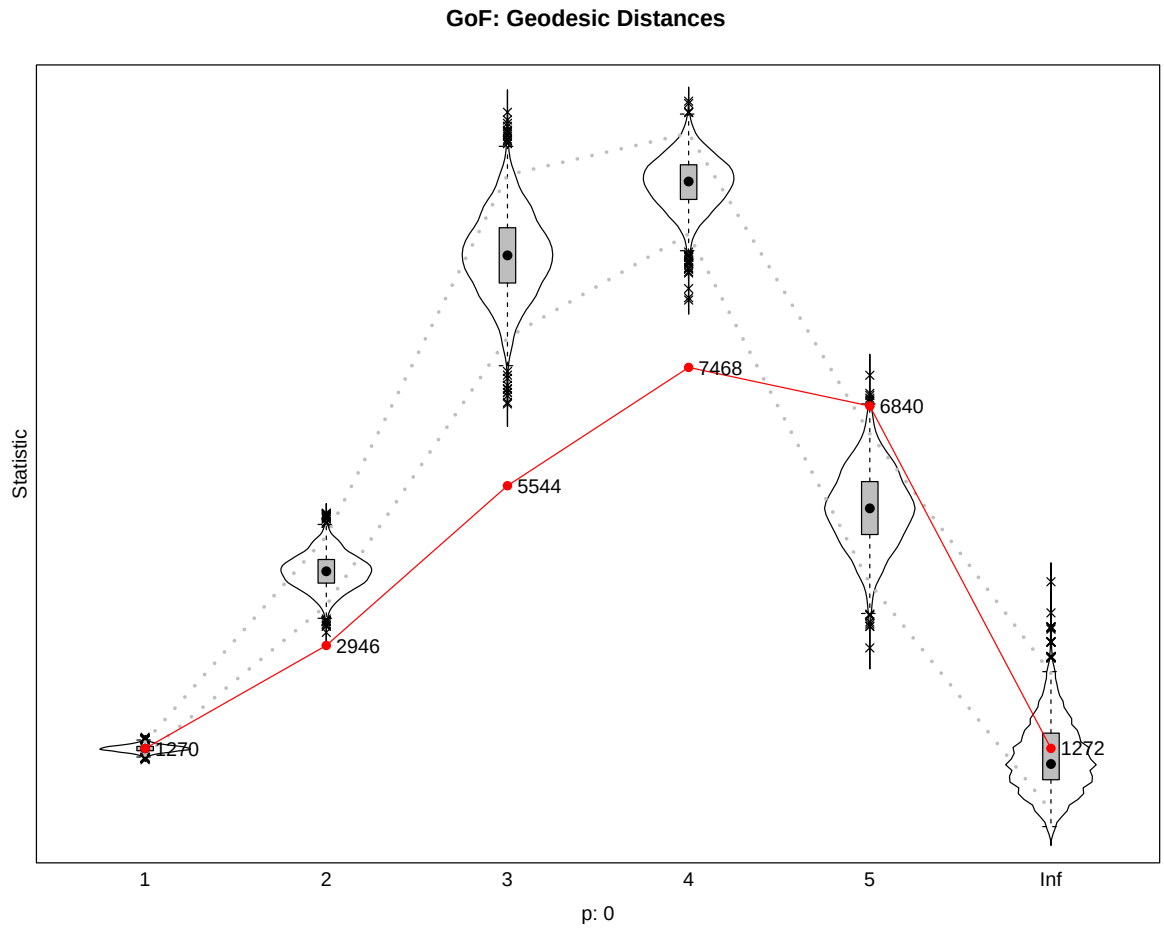
```
par(mfrow = c(2, 2))
plot(gof1_indeg,  main = "GoF: Indegree Distribution")
```



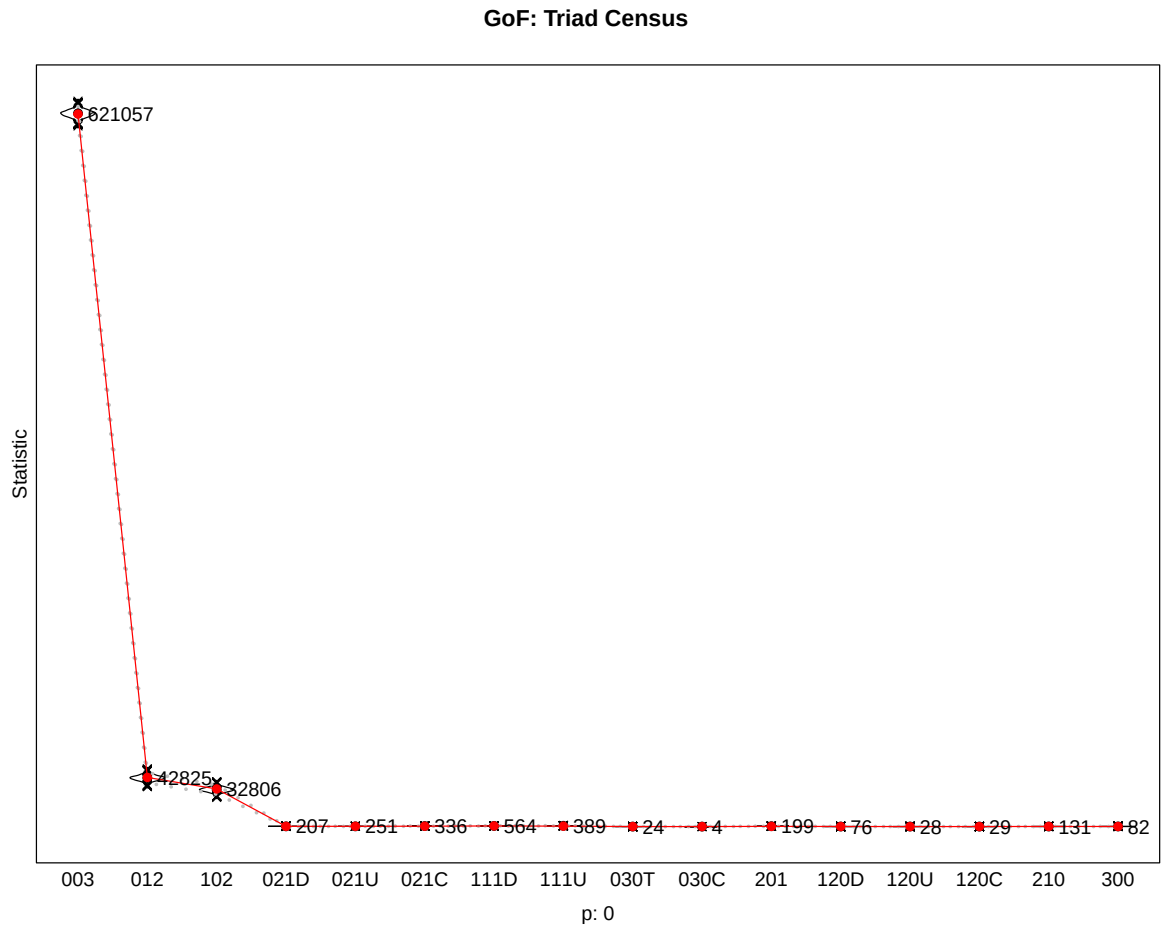

```
plot(gof1_outdeg, main = "GoF: Outdegree Distribution")
```




```
plot(gof1_geo,    main = "GoF: Geodesic Distances")
```




```
plot(gof1_tri,    main = "GoF: Triad Census")
```



Reading the table: a p-value above 0.05 means the simulated and observed distributions are not distinguishable (i.e. the model fits that statistic).

- **Indegree** fits well (p = 0.667). This is consistent with `inPopSqrt` directly constraining the indegree shape.
- **Outdegree, geodesics, triads**: all p 0.

The likely cause is the questionnaire's six-friend cap. The empirical outdegree distribution is right-truncated at 6, but the linear `outdegree (density)` term cannot reproduce that truncation, so the simulated networks have too much spread on outdegree. Geodesics and the triad census inherit the misfit. Task 3.3 proposes candidate fixes (`outAct`/`outActSqrt`, `gwespFF`/`transTies`).

Task 3.1.5 — Hypothesis Testing

Are hypotheses i.–iii. supported by the data?

```r
m1_results <- data.frame(
  Effect   = model1$effects$effectName,
  Estimate = round(model1$theta, 4),
  SE       = round(sqrt(diag(model1$covtheta)), 4),
  t_value  = round(model1$theta / sqrt(diag(model1$covtheta)), 3)
)
knitr::kable(m1_results, caption = "Model 1 - Estimates")
```

Table 5: Model 1 — Estimates

Effect	Estimate	SE	t_value
outdegree (density)	-2.7936	0.1774	-15.750
reciprocity	2.2083	0.1025	21.547
transitive triplets	0.6805	0.0438	15.544
3-cycles	-0.4594	0.0864	-5.320
indegree - popularity (sqrt)	-0.2995	0.0802	-3.735
logdistCov	-0.1525	0.0458	-3.326
genderCov alter	-0.1579	0.0978	-1.614
genderCov ego	0.1324	0.1024	1.293
same genderCov	0.8885	0.0944	9.416
alcoholCov similarity	0.3869	0.1407	2.749

H-i — popularity. Not supported. `inPopSqrt` is *negative* and significant ($\hat{\beta}$ = -0.3, t = -3.74). Once reciprocity and transitivity are in the model they absorb most of the "rich-get-richer" mass, and the residual popularity signal is slightly negative — popular pupils are not more likely to attract further ties.

H-ii — alcohol homophily. Supported. `alcoholCov similarity` is positive and significant ($\hat{\beta}$ = 0.387, t = 2.75). Pupils prefer to nominate peers with similar Wave-1 alcohol levels.

H-iii — living nearby. Supported. Log-distance has a negative significant effect ($\hat{\beta}$ = -0.152, t = -3.33). Greater distance lowers the probability of a friendship tie.

Other notable effects.

- Reciprocity ($\hat{\beta}$ = 2.21) and transitive triplets ($\hat{\beta}$ = 0.68) are strongly positive — both expected.
- 3-cycles is negative ($\hat{\beta}$ = -0.46), pointing to a hierarchical rather than generalised-exchange structure.
- Same-gender homophily is large ($\hat{\beta}$ = 0.89); the gender ego and alter main effects are not individually significant.

Task 3.2 — Co-evolution of Friendship and Alcohol

Task 3.2.1 — Model Specification (Selection + Influence)

Use the network specification from (1) for the selection part. Specify the influence part to test:
iv. Popular students tend to increase or maintain their level of alcohol consumption
v. Students tend to adjust their alcohol consumption to that of their friends

```
friendship2 <- sienaDependent(array(c(f1, f2, f3), dim = c(n, n, 3)))
drinkingbeh <- sienaDependent(alcohol, type = "behavior")

genderCov2  <- coCovar(as.numeric(gender))
ageCov2     <- coCovar(as.numeric(age))
logdistCov2 <- coDyadCovar(logdistance)

mydata2 <- sienaDataCreate(friendship2, drinkingbeh, genderCov2, ageCov2, logdistCov2)

myeff2 <- getEffects(mydata2)

# --- Selection part - same as Model 1, but alcohol is now the behavioural DV ---
myeff2 <- includeEffects(myeff2, transTrip, cycle3)
```

```
  effectNumber effectName          shortName include fix   test  initialValue
1 24           transitive triplets transTrip TRUE    FALSE FALSE          0
2 44           3-cycles            cycle3    TRUE    FALSE FALSE          0
  parm
1 0
2 0
```

```
myeff2 <- includeEffects(myeff2, inPopSqrt)
```

```
  effectNumber effectName                   shortName include fix   test
1 87           indegree - popularity (sqrt) inPopSqrt TRUE    FALSE FALSE
  initialValue parm
1          0   0
```

```
myeff2 <- includeEffects(myeff2, egoX, altX, sameX, interaction1 = "genderCov2")
```

```
  effectNumber effectName       shortName include fix   test  initialValue parm
1 298          genderCov2 alter altX      TRUE    FALSE FALSE          0   0
2 313          genderCov2 ego   egoX      TRUE    FALSE FALSE          0   0
3 369          same genderCov2  sameX     TRUE    FALSE FALSE          0   0
```

```r
myeff2 <- includeEffects(myeff2, X,    interaction1 = "logdistCov2")
```

```
  effectNumber effectName  shortName include fix   test  initialValue parm
1 265          logdistCov2 X         TRUE    FALSE FALSE          0   0
```

```r
myeff2 <- includeEffects(myeff2, simX, interaction1 = "drinkingbeh")
```

```
  effectNumber effectName             shortName include fix   test
1 727          drinkingbeh similarity simX      TRUE    FALSE FALSE
  initialValue parm
1          0   0
```

```r
# --- Influence part ---
# Hypothesis iv - popular students change their drinking
myeff2 <- includeEffects(myeff2, indeg,
                         name = "drinkingbeh", interaction1 = "friendship2")
```

```
  effectNumber effectName           shortName include fix   test  initialValue
1 980          drinkingbeh indegree indeg     TRUE    FALSE FALSE          0
  parm
1 0
```

```r
# Hypothesis v - students adjust to the average of their friends
myeff2 <- includeEffects(myeff2, avSim,
                         name = "drinkingbeh", interaction1 = "friendship2")
```

```
  effectNumber effectName                     shortName include fix   test
1 943          drinkingbeh average similarity avSim     TRUE    FALSE FALSE
  initialValue parm
1          0   0
```

```r
# Behavioural control - gender main effect on drinking
myeff2 <- includeEffects(myeff2, effFrom,
                         name = "drinkingbeh", interaction1 = "genderCov2")
```

```
  effectNumber effectName                          shortName include fix
1 1074         drinkingbeh: effect from genderCov2 effFrom   TRUE    FALSE
  test  initialValue parm
1 FALSE          0   0
```

The selection block mirrors Model 1, with one change: alcohol homophily now uses the *dynamic* behavioural variable (`simX` on `drinkingbeh`) instead of the Wave-1 snapshot.

The behaviour function already includes the linear and quadratic shape by default. On top of those:

- `indeg` for H-iv (popularity → drinking)
- `avSim` for H-v (friends' drinking → ego's drinking)
- `effFrom(genderCov2)` as a control on the behaviour

Task 3.2.2 — Estimation and Convergence

```
algo2 <- sienaAlgorithmCreate(projname = "saom_t32", seed = 161, n3 = 3000)
```

```
If you use this algorithm object, siena07 will create/use an output file saom_t32.txt .
```

```
if (file.exists("model2.RData")) {
  load("model2.RData")
} else {
  model2 <- siena07(algo2, data = mydata2, effects = myeff2,
                    returnDeps = TRUE, batch = TRUE, verbose = FALSE)
  tries <- 0
  while ((model2$tconv.max > 0.25 || max(abs(model2$tconv)) > 0.10) && tries < 3) {
    tries <- tries + 1
    model2 <- siena07(algo2, data = mydata2, effects = myeff2,
                      prevAns = model2, returnDeps = TRUE,
                      batch = TRUE, verbose = FALSE)
  }
  save(model2, file = "model2.RData")
}
```

```
m2_conv <- data.frame(
  Statistic = c("Overall maximum convergence ratio (tconv.max)",
                "Maximum |t-ratio| per parameter"),
  Value     = c(round(model2$tconv.max, 3),
                round(max(abs(model2$tconv)), 3)),
  Threshold = c("< 0.25", "< 0.10")
)
knitr::kable(m2_conv, caption = "Model 2 - Convergence Diagnostics")
```

Table 6: Model 2 — Convergence Diagnostics

Statistic	Value	Threshold
Overall maximum convergence ratio (tconv.max)	0.178	< 0.25
Maximum |t-ratio| per parameter	0.055	< 0.10

`tconv.max` = 0.178, max |t| per parameter = 0.055. Both clear the 0.25 / 0.10 thresholds, so the joint network–behaviour model converged.

`tconv.max` is higher than in Model 1 (0.091), but that is expected: Model 2 has 19 parameters instead of 10 and adds a behavioural dependent variable, which makes the moment-matching problem stiffer.

Task 3.2.3 — Goodness of Fit

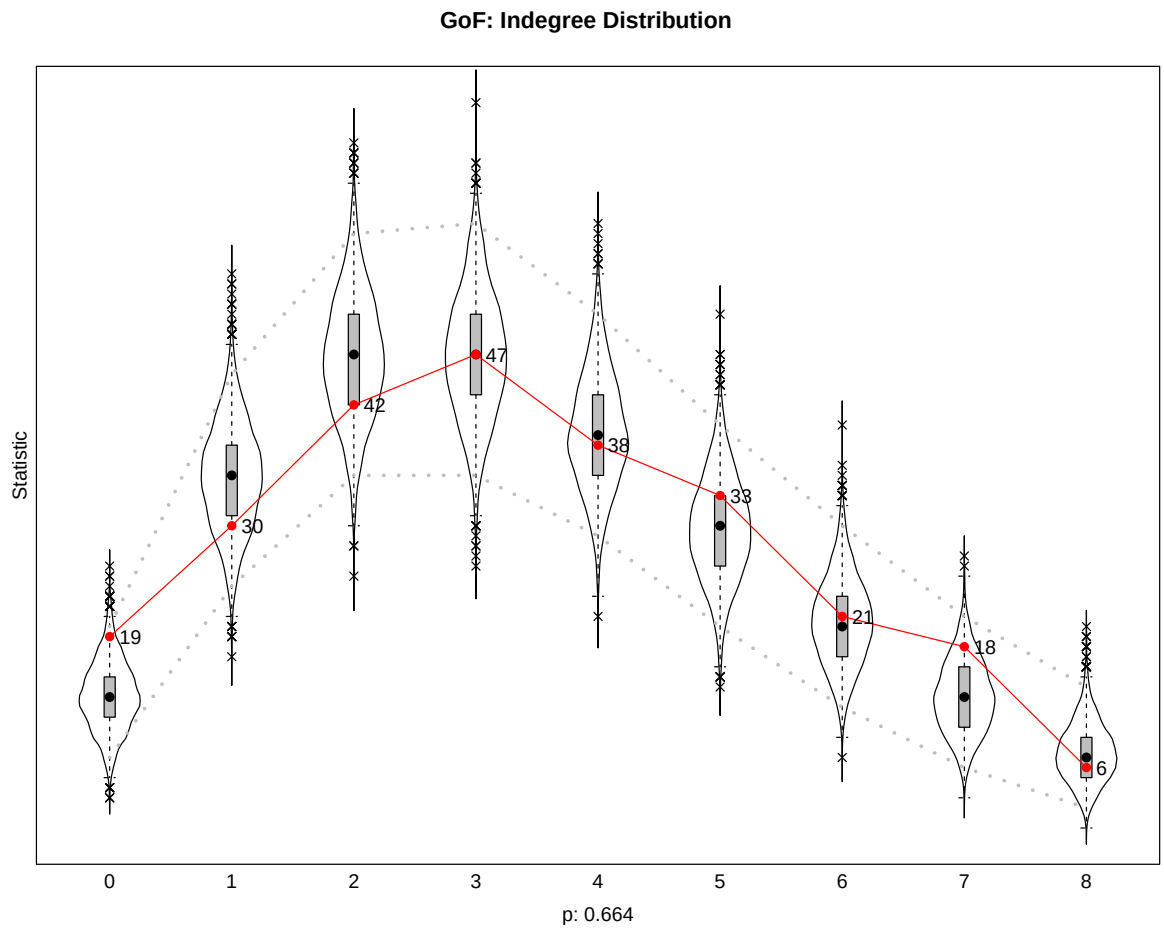
```
if (file.exists("gof2.RData")) {
  load("gof2.RData")
} else {
  gof2_indeg  <- sienaGOF(model2, IndegreeDistribution,  varName = "friendship2", cumulative
  gof2_outdeg <- sienaGOF(model2, OutdegreeDistribution, varName = "friendship2", cumulative
  gof2_geo    <- sienaGOF(model2, GeodesicDistribution,  varName = "friendship2", cumulative
  gof2_beh    <- sienaGOF(model2, BehaviorDistribution,  varName = "drinkingbeh")
  save(gof2_indeg, gof2_outdeg, gof2_geo, gof2_beh, file = "gof2.RData")
}
```

```
gof2_p <- data.frame(
  Statistic = c("Indegree distribution", "Outdegree distribution",
                "Geodesic distribution", "Behaviour distribution"),
  `p-value` = c(round(gof2_indeg$Joint$p, 3),
                round(gof2_outdeg$Joint$p, 3),
                round(gof2_geo$Joint$p, 3),
                round(gof2_beh$Joint$p, 3)),
  check.names = FALSE
)
knitr::kable(gof2_p, caption = "Model 2 - sienaGOF p-values (Mahalanobis)")
```

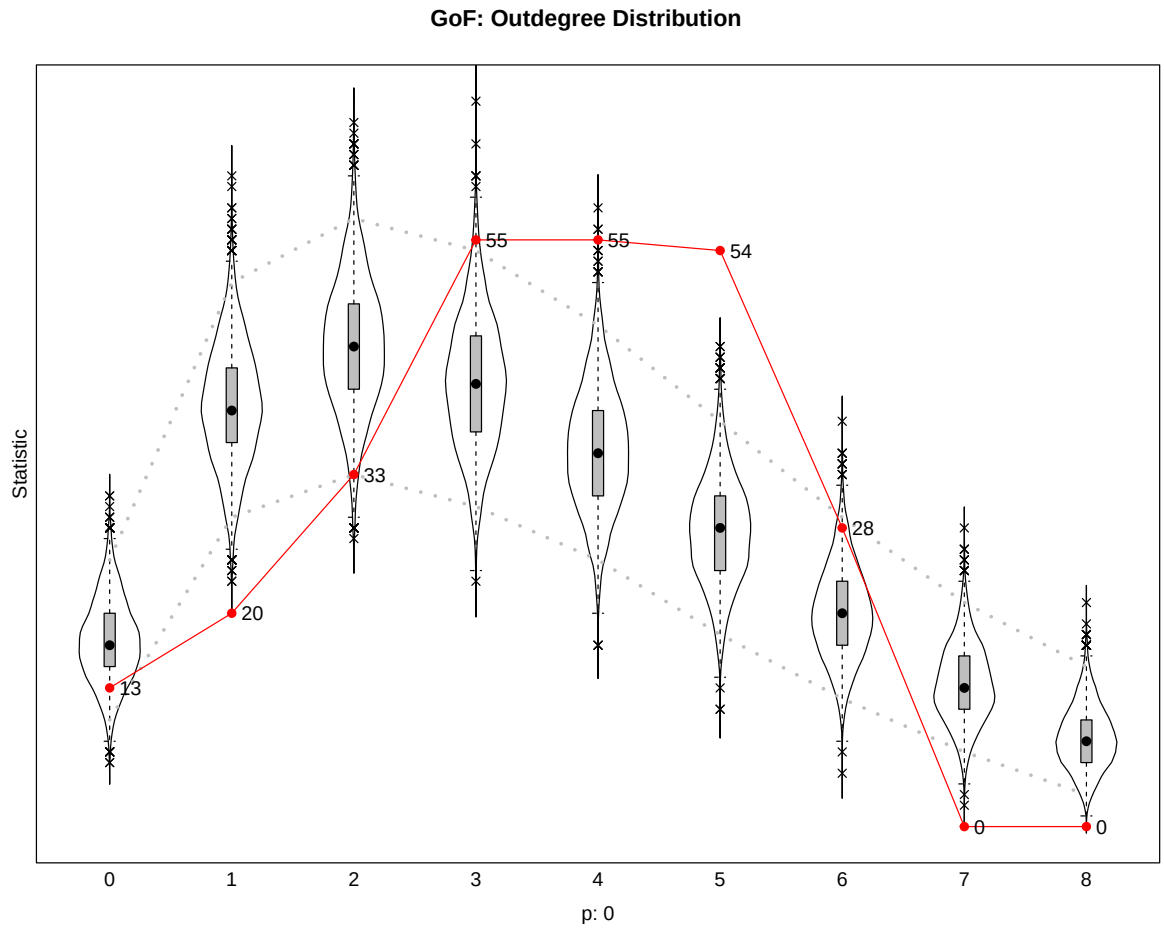
Table 7: Model 2 — sienaGOF p-values (Mahalanobis)

Statistic	p-value
Indegree distribution	0.664
Outdegree distribution	0.000
Geodesic distribution	0.000
Behaviour distribution	0.610

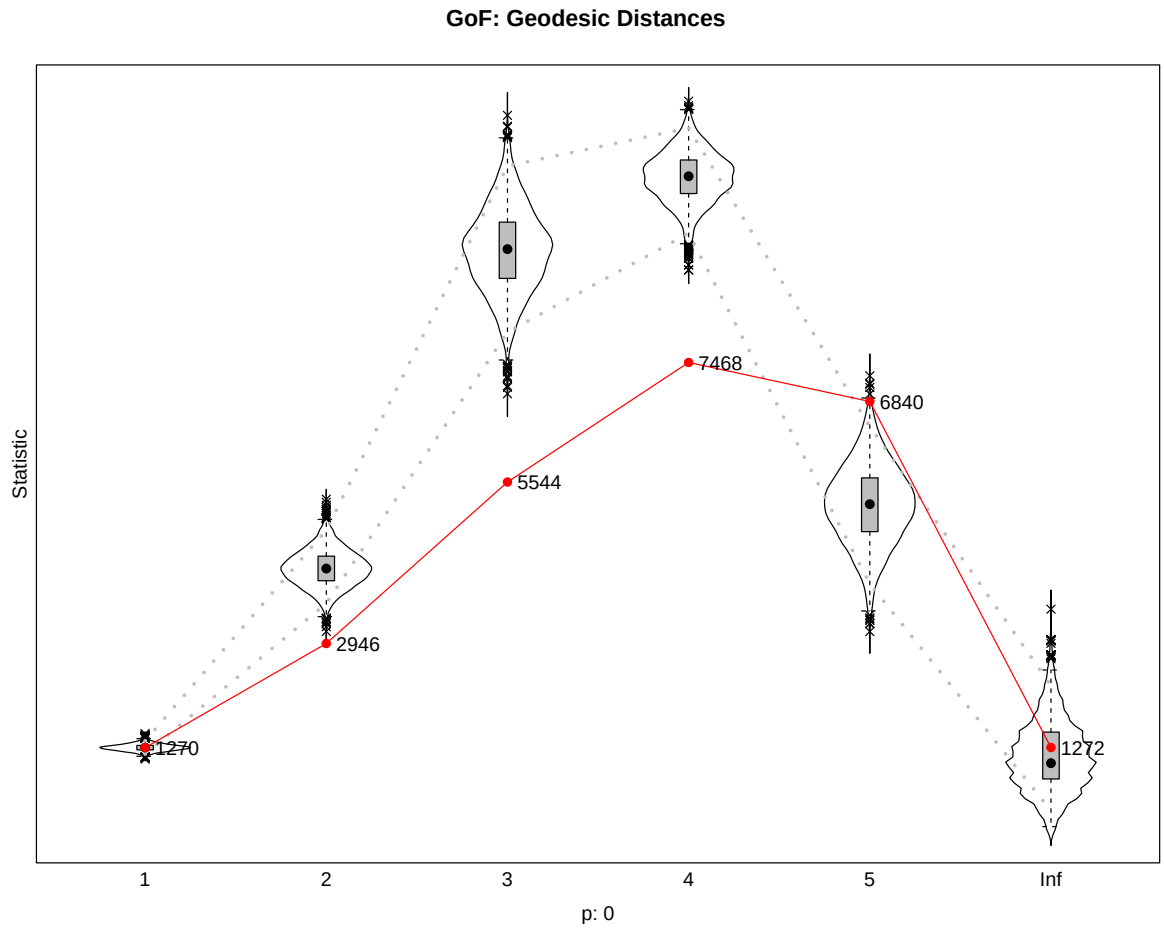
```
par(mfrow = c(2, 2))
plot(gof2_indeg,  main = "GoF: Indegree Distribution")
```




```
plot(gof2_outdeg, main = "GoF: Outdegree Distribution")
```

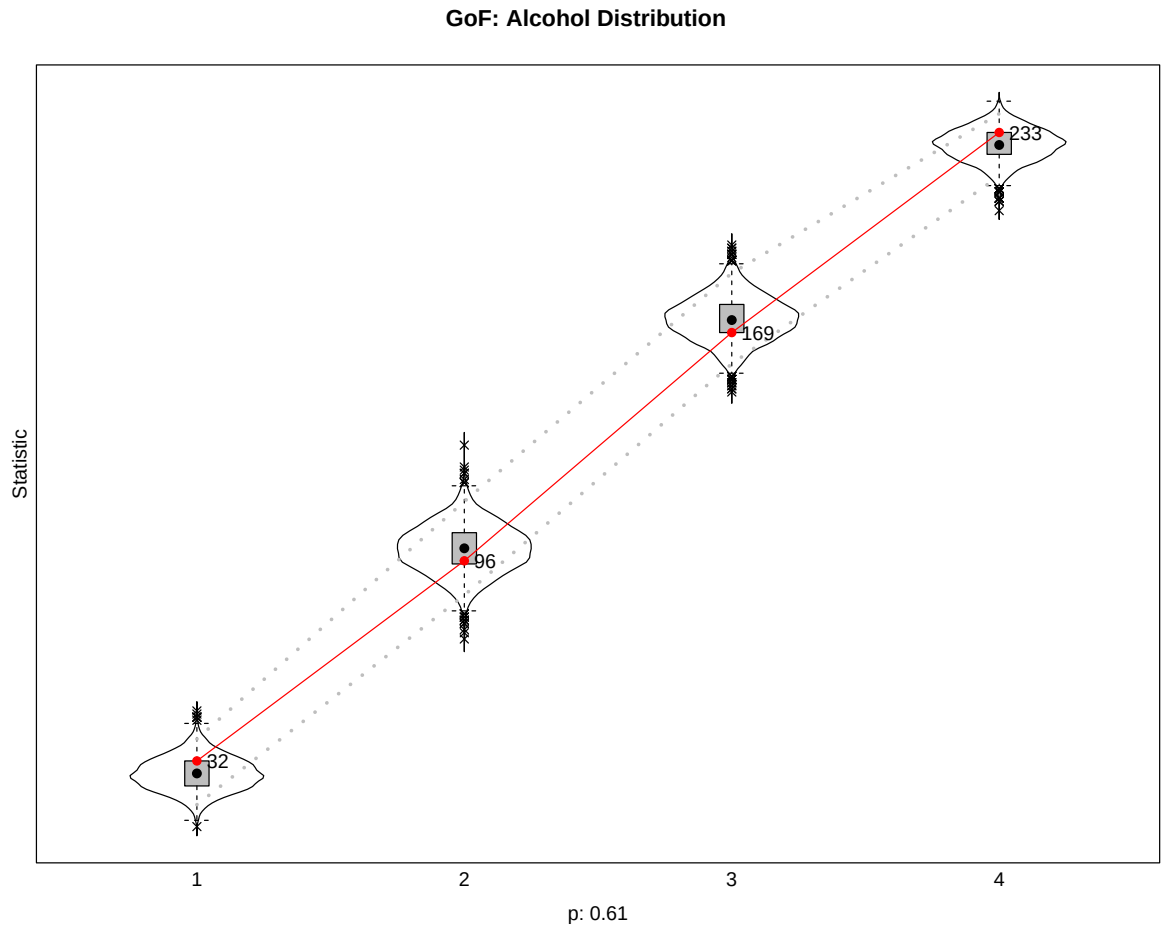



```
plot(gof2_geo,    main = "GoF: Geodesic Distances")
```




```r
plot(gof2_beh,    main = "GoF: Alcohol Distribution")
```

```
Note: some statistics are not plotted because their variance is 0.
This holds for the statistic: 5.
```



Same pattern as Model 1 on the network side, with one new fit to evaluate (the behaviour distribution):

- **Indegree**: fits (p = 0.664).
- **Behaviour distribution**: fits well (p = 0.61) — the alcohol side of the model is adequate.
- **Outdegree and geodesics**: still p 0, same six-nomination-cap issue. Task 3.3 covers candidate fixes.

Task 3.2.4 — Hypothesis Testing

Are hypotheses iv.–v. supported? How do conclusions about i.–iii. differ from (1)?

```
m2_results <- data.frame(
  Effect   = model2$effects$effectName,
  Estimate = round(model2$theta, 4),
```

```
  SE       = round(sqrt(diag(model2$covtheta)), 4),
  t_value  = round(model2$theta / sqrt(diag(model2$covtheta)), 3)
)
knitr::kable(m2_results, caption = "Model 2 - Estimates")
```

Table 8: Model 2 — Estimates

Effect	Estimate	SE	t_value
constant friendship2 rate (period 1)	10.9352	1.0565	10.351
constant friendship2 rate (period 2)	8.9980	0.8542	10.534
outdegree (density)	-2.8040	0.1873	-14.971
reciprocity	2.1664	0.1085	19.964
transitive triplets	0.6722	0.0450	14.922
3-cycles	-0.4499	0.0895	-5.027
indegree - popularity (sqrt)	-0.3027	0.0837	-3.617
logdistCov2	-0.1617	0.0460	-3.515
genderCov2 alter	-0.1543	0.1042	-1.480
genderCov2 ego	0.1412	0.1007	1.402
same genderCov2	0.8886	0.0967	9.192
drinkingbeh similarity	0.8923	0.2716	3.285
rate drinkingbeh (period 1)	1.5293	0.2447	6.251
rate drinkingbeh (period 2)	2.3287	0.4014	5.801
drinkingbeh linear shape	0.1828	0.2441	0.749
drinkingbeh quadratic shape	0.0644	0.0707	0.911
drinkingbeh average similarity	4.9856	1.5566	3.203
drinkingbeh indegree	0.0466	0.0639	0.729
drinkingbeh: effect from genderCov2	0.0044	0.1779	0.024

H-iv — popular students drink more. Not supported. `drinkingbeh indegree` is small and not significant ($\hat{\beta} = 0.047$, $t = 0.73$). Receiving many nominations does not push drinking up, at least once `avSim` (the influence channel) is in the model.

H-v — assimilation to friends' drinking. Supported. `drinkingbeh average similarity` is large and significant ($\hat{\beta} = 4.99$, $t = 3.2$). Pupils whose alcohol level differs from their friends' average move toward it over time.

Revisiting H-i to H-iii in the joint model. The three selection-side coefficients are close to Model 1:

- **H-i (popularity):** still negative and significant ($\hat{\beta}$ = -0.303, t = -3.62) — still **not supported**.

- **H-iii (living nearby):** still negative and significant on log-distance ($\hat{\beta}$ = -0.162, t = -3.52) — still **supported**.
- **H-ii (alcohol homophily):** now uses the evolving behaviour and is *stronger* than in Model 1 ($\hat{\beta} = 0.892$ vs 0.387; $t = 3.28$) — still **supported**.

The bigger homophily coefficient under the dynamic DV is plausible: in Model 1 the Wave-1 snapshot is a noisy proxy for "alcohol level at the time the tie formed", which attenuates the estimate. Using the dynamic DV removes that attenuation.

Task 3.2.5 — Selection vs. Influence

Given the model in (2.2), do we have evidence for selection only, influence only, or both?

Both mechanisms show up in the model:

- **Selection** (`simX(drinkingbeh)`): $\hat{\beta} = 0.89$, $t = 3.28$.
- **Influence** (`avSim` on drinking): $\hat{\beta} = 4.99$, $t = 3.2$.

So we find evidence for **both selection and influence**. Pupils both pick friends whose drinking is similar to theirs (selection) *and* shift their own drinking toward their friends' average over time (influence).

This is the case where SAOMs are useful: cross-sectional homophily on alcohol is consistent with either mechanism alone, so without the joint dynamic model we could not separate them. The two effects are identifiable jointly because they act on different parts of the data — selection on ties given attributes, influence on attribute changes given ties.

Task 3.3 — Model Improvement

Discuss new effects that could improve the fit on geodesic distances and degree distributions.

The fit problems sit on three statistics: outdegree, geodesics, and (in Model 1) the triad census. They share a root cause — the questionnaire's six-friend cap, which a linear `outdegree (density)` term cannot reproduce. Once the outdegree distribution is wrong, geodesics and the triad census inherit the misfit.

Outdegree shape. Add `outActSqrt` (or `outAct`, or a `gwodegree`-style term). With a negative coefficient these saturate the propensity to send another tie, mimicking the cap. An `outdegree truncated at 6` term would do something similar by hard-coding the truncation point.

Indegree shape. `inActSqrt` lets the rate of receiving further nominations bend non-linearly with current indegree, which is more flexible than the strictly linear assumption built into `inPopSqrt`.

Triadic and geodesic fit. Replace (or supplement) `transTrip` with `gwespFF` or `transTies`. `transTrip` over-rewards ties that close *many* transitive paths simultaneously; `gwespFF` flattens that and tends to match the observed clustering distribution more reliably.

Other directions.

- `balance` or `transRecTrip` to capture reciprocity-transitivity interactions, which are part of the triad census the current model misses.
- An `absDiffX` on age (a dyadic age-difference covariate) — friendships within cohorts often cluster, and tighter clusters tend to improve the geodesic fit.

Task 3.4 — Two Additional Hypotheses

Two further testable hypotheses on friendship/alcohol dynamics, and how to operationalise them.

Hypothesis A — Drinking and smoking co-evolve. Adolescent alcohol and tobacco use are often correlated. Beyond the network-mediated channels, the two behaviours may push each other directly.

Operationalisation. Add smoking as a second behavioural DV (`sienaDependent(smoke, type = "behavior")`) and include the cross-behaviour effects `effFrom(drinkingbeh, smoke)` and `effFrom(smoke, drinkingbeh)`. A positive cross-effect would mean drinking pushes up smoking and vice versa, on top of the network-driven influence and selection. Requires per-wave smoking data; otherwise a standard SAOM extension.

Hypothesis B — Heavy drinkers have less stable friendships. Alcohol use may shape not only *who* a pupil is friends with, but also how stable those friendships are: heavier drinkers may churn through ties faster.

Operationalisation. Either let the network rate depend on alcohol via `effFrom(rate, drinkingbeh)`, or split `egoX(drinkingbeh)` into its `creation` and `endowment` (tie-maintenance) variants. A *negative endowment* effect (or *positive rate* effect of drinking) would mean heavy drinkers are more likely to drop existing ties, separating tie *formation* from tie *persistence*.

Reproducibility Footer

```r
cat("Seed used: 161\n")
```

```
Seed used: 161
```

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.7.4

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Zurich
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] sna_2.8               network_1.20.0        statnet.common_4.13.0
[4] RSiena_1.6.6

loaded via a namespace (and not attached):
 [1] vctrs_0.7.1      cli_3.6.5        knitr_1.51       rlang_1.1.7
 [5] xfun_0.56        jsonlite_2.0.0   xtable_1.8-8     glue_1.8.0
 [9] htmltools_0.5.9  rmarkdown_2.30   grid_4.5.2       evaluate_1.0.5
[13] tibble_3.3.1     MASS_7.3-65      fastmap_1.2.0    yaml_2.3.12
[17] lifecycle_1.0.5  compiler_4.5.2   codetools_0.2-20 coda_0.19-4.1
[21] pkgconfig_2.0.3  lattice_0.22-7   digest_0.6.39    pillar_1.11.1
[25] parallel_4.5.2   magrittr_2.0.4   Matrix_1.7-4     tools_4.5.2
```